

BookBot Bin Logic Challenge

Retrieve by address, not by conveyor race

At a glance: Retrieve the right book by using bin addresses, not a conveyor race. Plan about 80 minutes for the core block; 4 to 6 teams of 4 to 5 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: automated storage and retrieval systems, bin addressing, routing rules, algorithmic search (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 25 STUDENTS AND 6 TEAMS

- bin cards
- address labels
- order deck
- planning route mats: one per team; shared judging mat: one
- timers
- optional grabber tool

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Give each team a planning copy, then stage one shared tabletop judging mat. Mark DEPOT immediately left of A1 and run one team at a time.
- 02 Stage the scoring sheet and any reference values before testing begins.
- 03 Use tabletop pointers only. Keep the shared judging mat clear of actual library operations; no gloves, goggles, or trays are required.
- 04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Retrieve the right book by using bin addresses, not a conveyor race." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the route rules: start and finish at DEPOT; one move is an orthogonal step between adjacent bins or between DEPOT and A1; retrieve every requested address; no diagonal moves or jumps.

Phase 4 Build or solve

0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Test and record

Teams plan on their own copies, then run one at a time on the shared mat. Record legal-move count before time; a missed address or illegal jump makes the route incomplete. Use the instructor route and scoring key to compare the card minimum and normalize the efficiency score.

Phase 6 Redesign

Each team changes exactly one variable, predicts the effect, and re-tests to compare against baseline.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, clear their pointer and order card, and reset the shared mat to DEPOT.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: No measurable result on the first test. **Fix:** Check the setup against the steps, confirm the standard test was followed, and log the baseline anyway.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to automated storage and retrieval.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Why store by address instead of by subject?
- What made one route faster than another?
- Where else is this storage logic used?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Correct retrievals	35
Legal complete route	25
Routing or algorithm explanation	20
Efficiency versus card optimum	10
Teamwork	10
Total	100

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES Temple Charles Library BookBot; Charles Library Circulating Collections